



CS540 Introduction to Artificial Intelligence

Ethics and Trust in AI

University of Wisconsin-Madison
Spring 2025

Announcements

- **Homework:**
 - HW10 due on Friday **May 2nd at 11:59 PM**
- **Final Exam Information: next slide**
- **Course evaluation**

Final Information

- **Time: May 7th 07:45 AM - 09:45 AM**
- **Location (by section**):**
 - **Lecture 001 (Instructor Sharon Li): S429 Chemistry Building**
 - **Lecture 002 (Instructor Fred Sala): 1220 Microbial Sciences**
 - **Lecture 003 (Instructor Blerina Gkotse): B10 Ingraham Hall**

****To find your section go to MyUW->Course Schedule->It will say "LEC 00_". Do not use canvas to find your section (everyone will see CS540 001 since we merged the canvas site for all three sections).**

- **Format:** The final exam will be entirely multiple choice.
- **Cheat Sheet:** You will be allowed a cheat sheet of a single piece of paper (8.5" x 11", front and back). The exam will focus on conceptual and applied AI reasoning.
- **Calculator:** Calculators are allowed if they don't have an internet connection. A calculator will not be necessary though it may be useful to double check simple arithmetic.
- **Detailed topic list + practice:** <https://piazza.com/class/m5zvrf0clyo3sl/post/449>

I am

Good

Evil

Yes

Lucky for the world

AI dual use:

- VX chemical compound
- deep fake
- Autonomous weapons
- ...

I know AI

No

Beware AI lacks:

- Fairness
- Privacy
- Explainability
- Trust
- ...

Lucky for the world

Dual use of artificial-intelligence-powered drug discovery

- Key observation: flip the objective function to make optimization find many highly toxic compounds

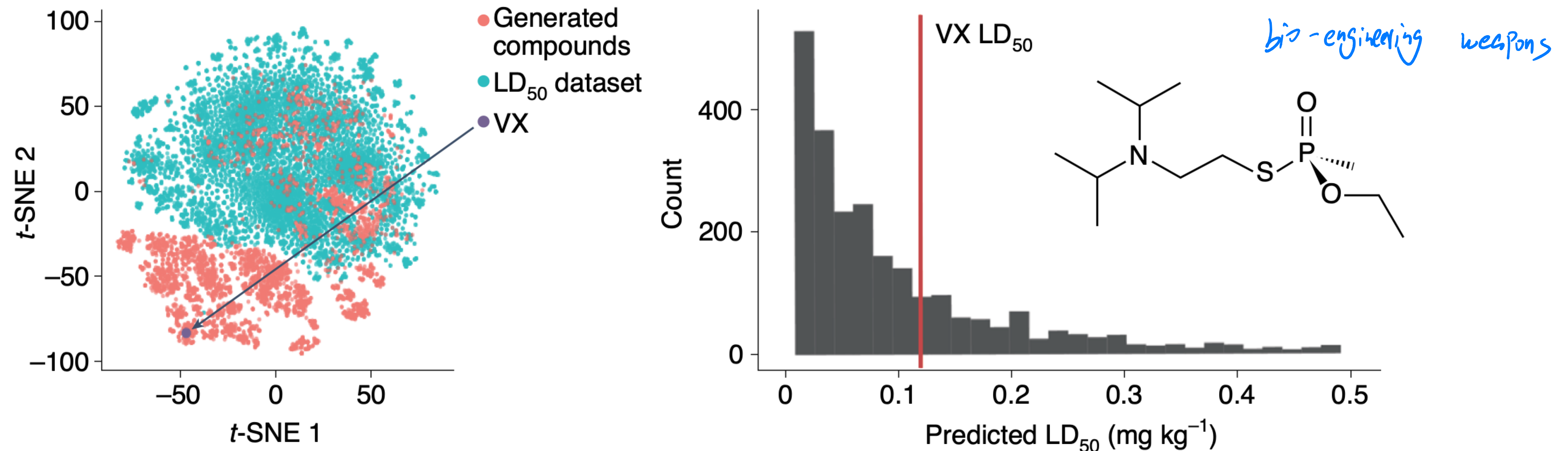


Fig. 1 | A t-SNE plot visualization of the LD₅₀ dataset and top 2,000 MegaSyn AI-generated and predicted toxic molecules illustrating VX. Many of the molecules generated are predicted to be more toxic in vivo in the animal model than VX (histogram at right shows cut-off for VX LD₅₀). The 2D chemical structure of VX is shown on the right.

[Urbina et al. Nature machine intelligence 2022]

<https://www.youtube.com/watch?v=cQ54GDm1eL0>

Example 1: Fake Obama Video



Example 2: Fake face Images by GAN

- Which are real/fake? <https://thispersondoesnotexist.com/>



Example 3: fiction Generated by GPT-3

- Completing a prompt from “Harry Potter and the Methods of Rationality”:

“... If there were any other monster that could defeat you as easily as that one, then you would have died of it long ago. That monster is stupidity. And that is why, my young apprentices, you must never, never, NEVER use the Killing Curse on anything without a brain!"

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“The Killing Curse is too good for something without a brain. You will be fighting brains, or something near enough that makes no real difference. You will not be fighting trolls. You will not be fighting Dementors. The Killing Curse is no tool for anything less than the third most perfect killing machine in all Nature. If you are not prepared to use it against a mountain troll, then you are not prepared to use it at all. Now. Pay attention to yourselves as I cast a simple spell. Listen to your own thoughts as I tell you how stupid you are.”

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Lucky for the world

Outline

- Bias and Fairness
- Fake Content
- Privacy
- Adversarial robustness



Bias and Fairness

Example

- US doctors: 60% male, 40% female
- AI: “Appointment with your doctor at 8am; ___ asks you to arrive early.” (He/She)?
 - Assume AI doesn’t know the doctor.
- $P(y = M) = 0.6$, $P(y = F) = 1 - P(y = M) = 0.4$
- Bayes optimal prediction: $\hat{y} = \underset{y}{\operatorname{argmax}} P(y) = M$
- Optimal error rate $P(\hat{y} \neq y) = P(y \neq M) = 0.4$.
- Potential harm: AI never addresses a doctor by “She”.
 - Biased? Sexist?

Example

- What is more fair?
- How about $P(\hat{y} = M \mid y = M) = P(\hat{y} = F \mid y = F)$
 - I.e., Probability of correct response same for men and women.
- But AI does not know y .
- Can achieve above by randomization: regardless of the actual doctor, predict M or F with probability 0.5
- More fair now (?), but suffer in error rate

$$P(\hat{y} \neq y) = P(y \neq M \mid y = M)P(y = M) + P(y \neq F \mid y = F)P(y = F) = 0.5$$

$\underbrace{\hspace{1.5cm}}_{0.5}$ $\underbrace{\hspace{1.5cm}}_{0.6}$ $\underbrace{\hspace{1.5cm}}_{0.5}$ $\underbrace{\hspace{1.5cm}}_{0.4}$

Example 2: Skin color bias in face recognition



<https://www.nytimes.com/2020/11/11/movies/coded-bias-review.html>

Example 3: Gender Bias in GPT-3

- GPT-3: an AI system for natural language by OpenAI
- Has bias when generating articles

Table 6.1: Most Biased Descriptive Words in 175B Model

Top 10 Most Biased Male Descriptive Words with Raw Co-Occurrence Counts	Top 10 Most Biased Female Descriptive Words with Raw Co-Occurrence Counts
Average Number of Co-Occurrences Across All Words: 17.5	Average Number of Co-Occurrences Across All Words: 23.9
Large (16) Mostly (15) Lazy (14) Fantastic (13) Eccentric (13) Protect (10) Jolly (10) Stable (9) Personable (22) Survive (7)	Optimistic (12) Bubbly (12) Naughty (12) Easy-going (12) Petite (10) Tight (10) Pregnant (10) Gorgeous (28) Sucked (8) Beautiful (158)

What causes bias in ML?

- Spurious correlation
 - e.g. the relationship between “man” and “computer programmers” was found to be highly similar to that between “woman” and “homemaker” ([Bolukbasi et al. 2016](#))
- Sample size disparity
 - If the training data coming from the minority group is much less than those coming from the majority group, it is less likely to model well the minority group.
- Proxies
 - Even if sensitive attribute (attributes that are considered should not be used for a task e.g. race/gender) is not used for training a ML system, there can always be other features that are proxies of the sensitive attribute(e.g. neighborhood).

How to mitigate bias?

- **Removing bias from data**
 - Collect representative data from minority groups
 - Remove bias associations
- **Designing fair learning methods**
 - Add fairness constraints to the optimization problem for learning

Group fairness

$y \in \{0,1\}$: true label (Example: loan eligibility)

$\hat{y} \in \{0,1\}$: predicted label (Example: AI recommends loan)

$G \in \{1 \dots, K\}$: sensitive groups

Demographic parity: (Regardless with the group you are in, the probability of eligible for loan is the same)

$$P(\hat{y} = 1 \mid G = 1) = \dots = P(\hat{y} = 1 \mid G = K)$$

Equal opportunity:

$$P(\hat{y} = 1 \mid G = 1, y = 1) = \dots = P(\hat{y} = 1 \mid G = K, y = 1)$$

→ The chance that the system makes an error is the same across all groups



Fake Content and Misinformation

<https://www.youtube.com/watch?v=cQ54GDm1eL0>

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Detecting Fake Content

Fake photos/videos can have drawbacks.





Privacy

Example 1: Netflix Prize Competition

- Netflix Dataset: 480189 users x 17770 movies



	movie 1	movie 2	movie 3	movie 4	movie 5	movie 6
Tom	5	?	?	1	3	?
George	?	?	3	1	2	5
Susan	4	3	1	?	5	1
Beth	4	3	?	2	4	2

- The data was released by Netflix in 2006
 - replaced individual names with random numbers
 - moved around personal details, etc

Example 1: Netflix Prize Competition

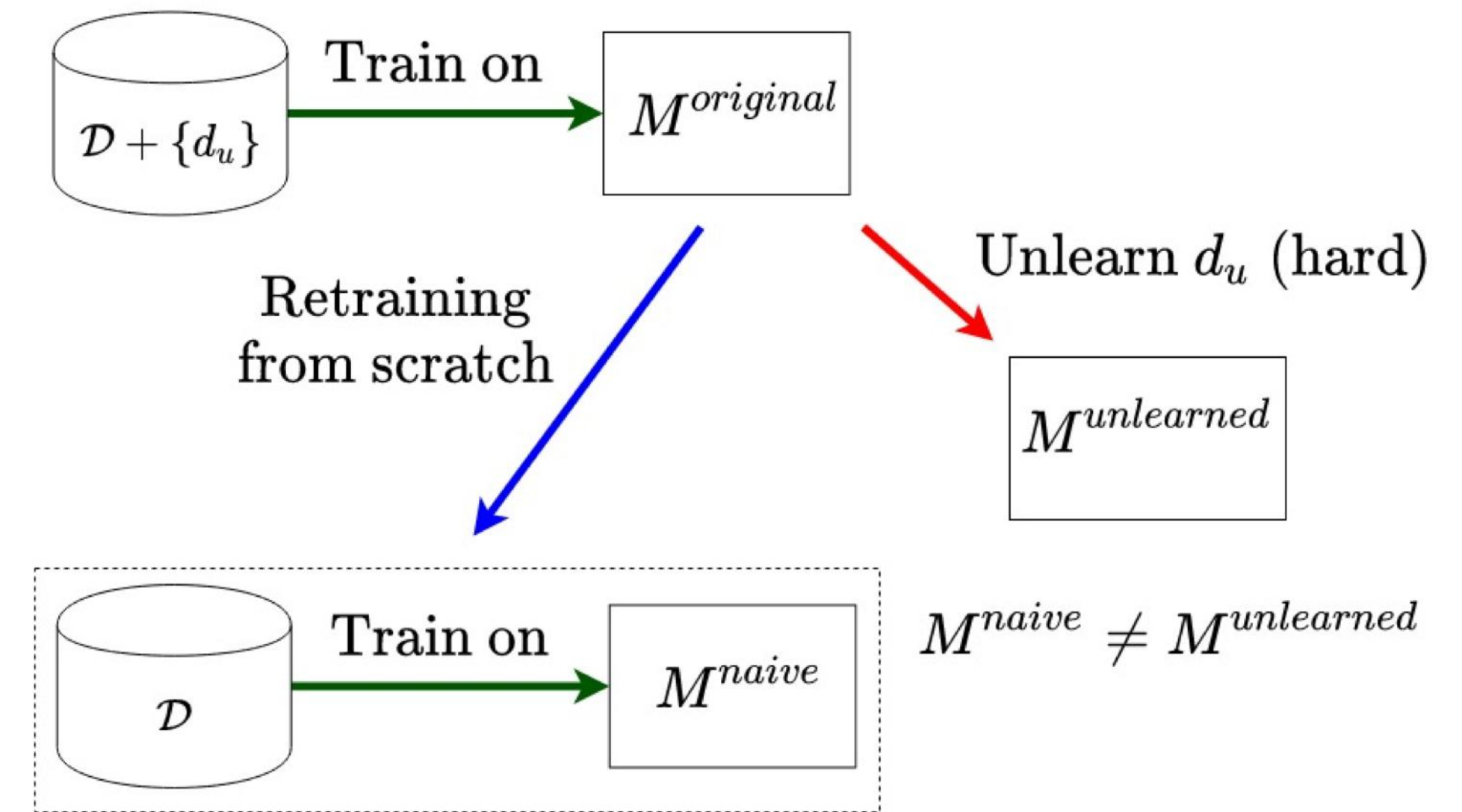
- [Arvind Narayanan](#) and [Vitaly Shmatikov](#) compared the data with the non-anonymous IMDb users' movie ratings
- Very little information from the database was needed to identify the subscriber
 - simply knowing data about only two movies a user has reviewed allows for 68% re-identification success

Right to be Forgotten

- The right to request that personally identifiable data be deleted
- E.g., an individual who did something foolish as a teenager doesn't want it to appear in web searches for the name for the rest of the life

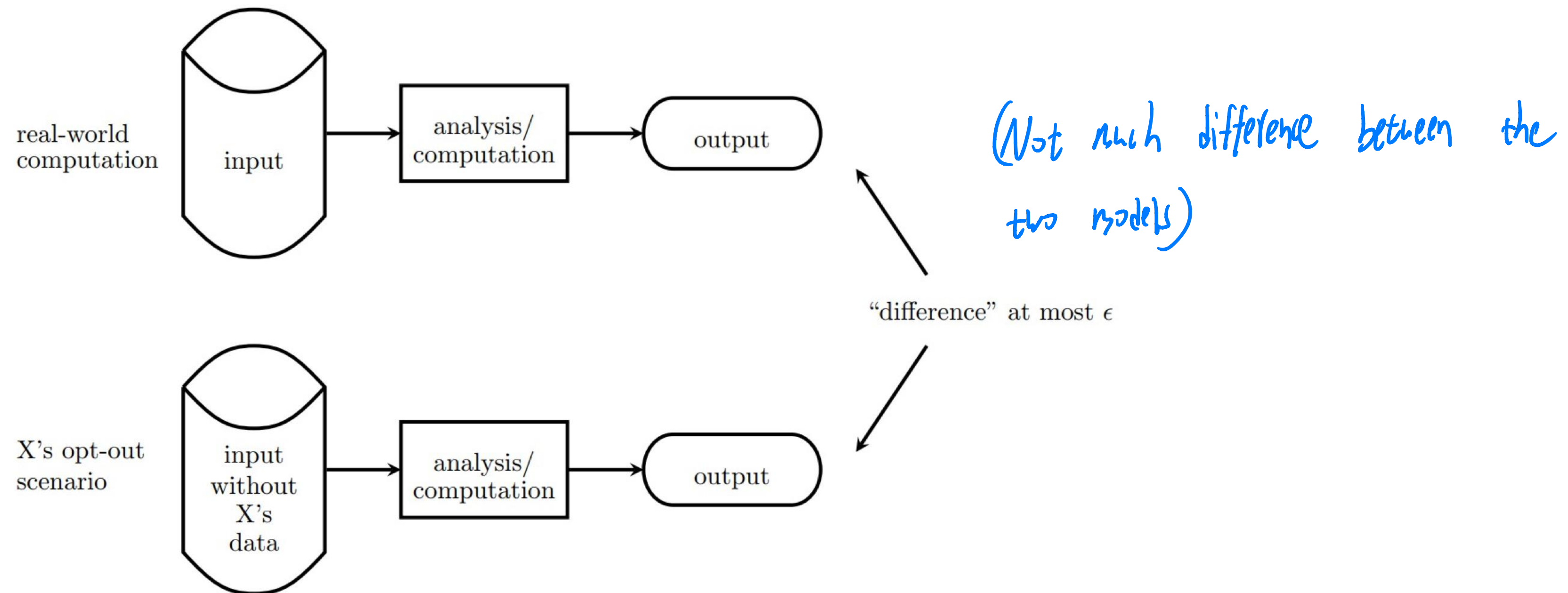
Right to be Forgotten

- What if the data has been used in training a deep network?
 - Need to **unlearn**
- Other issues
 - Multiple copies of the data
 - Data already shared with others



Popular framework: Differential Privacy

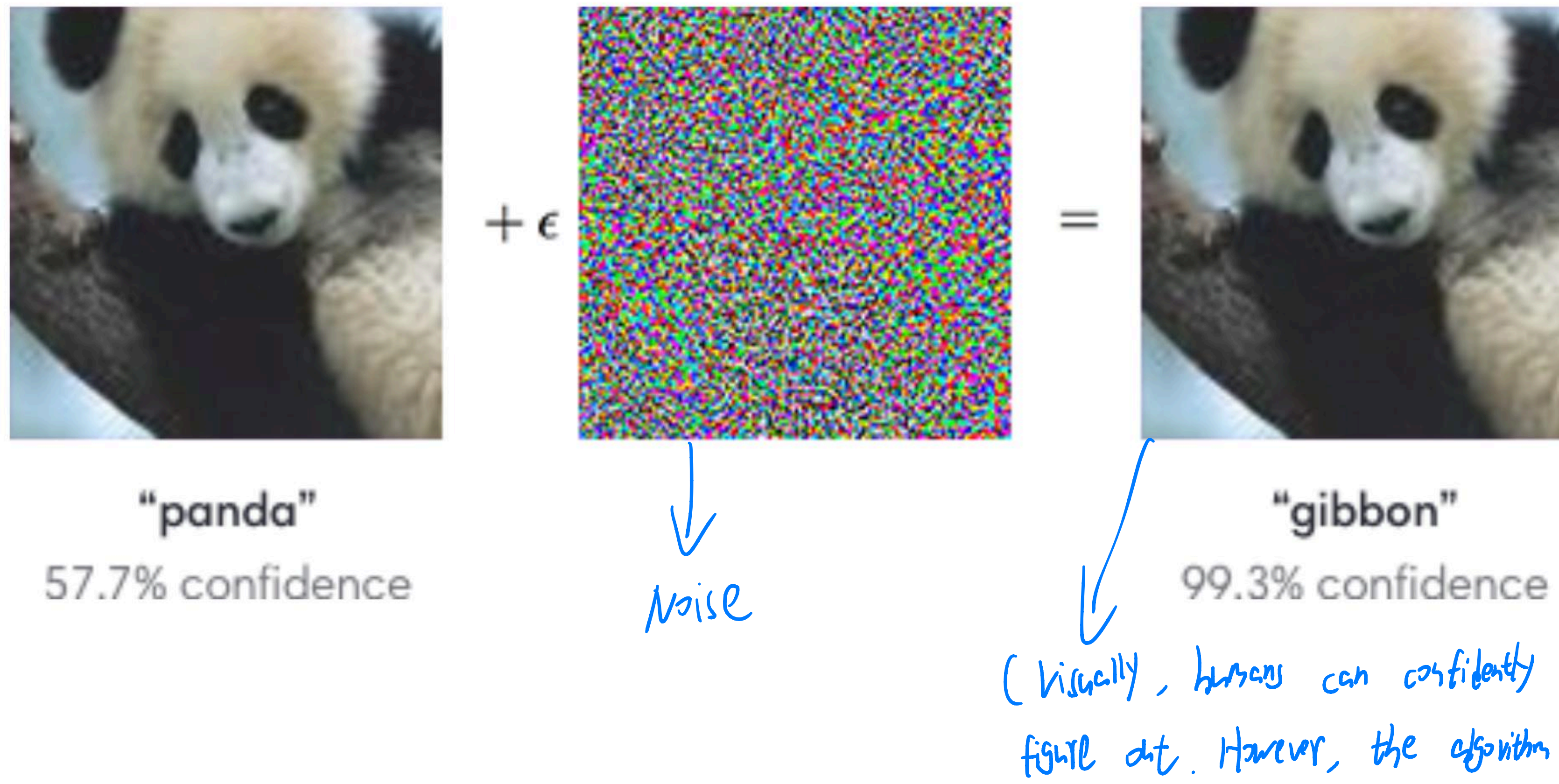
- The computation is differential private, if removing any data point from the dataset will only change the output very slightly
- Usually done by adding noise to the dataset





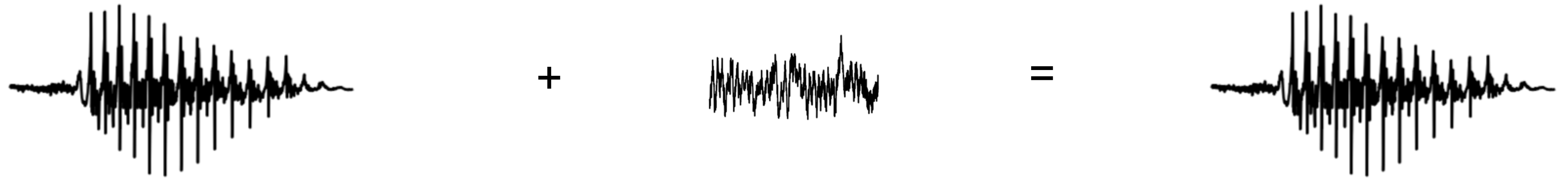
Robustness in AI

Manipulate Classification



<https://openai.com/blog/adversarial-example-research/>

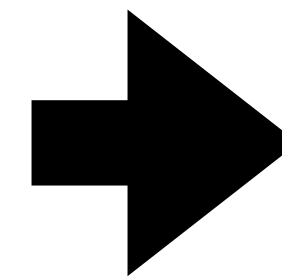
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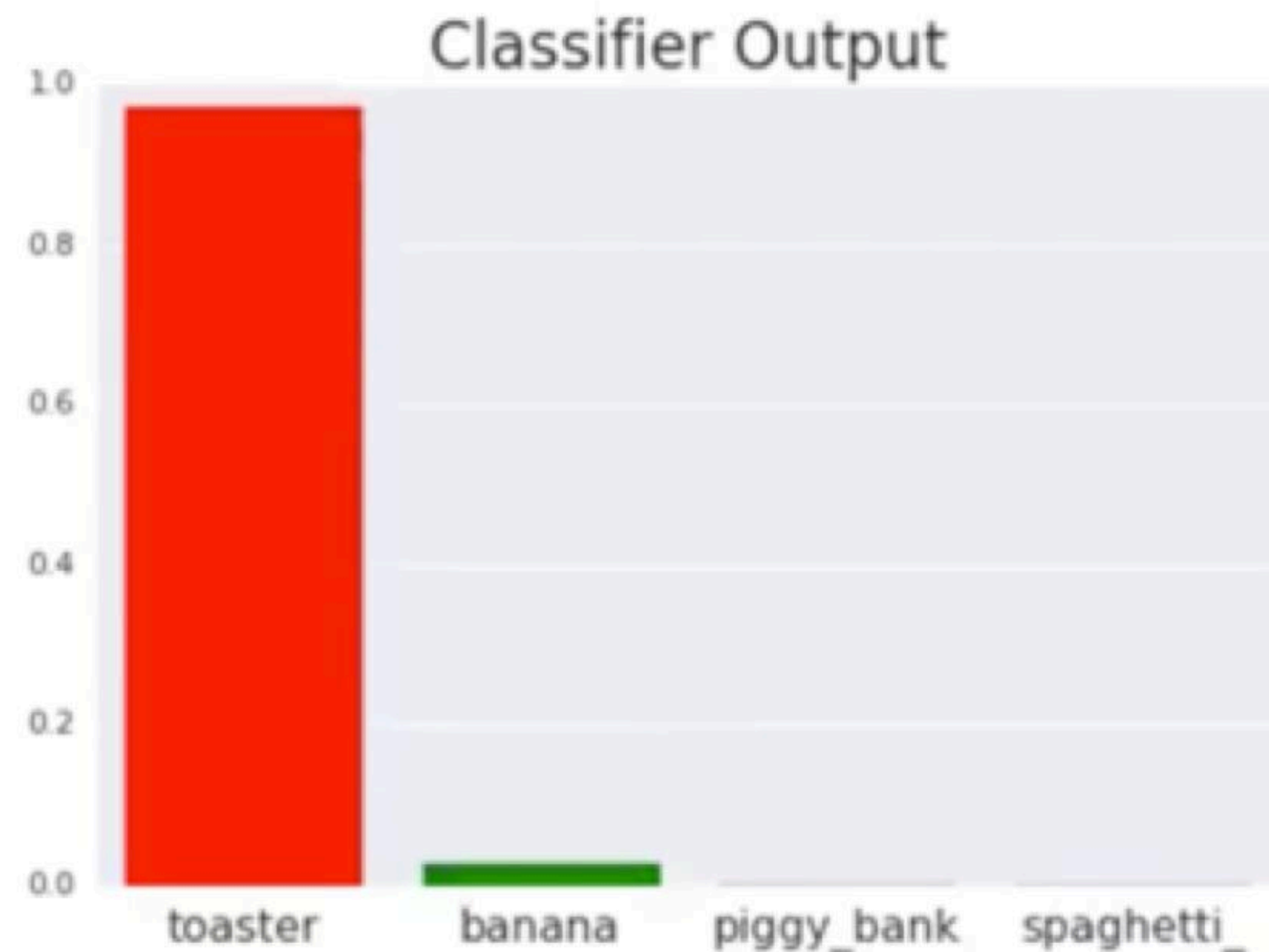
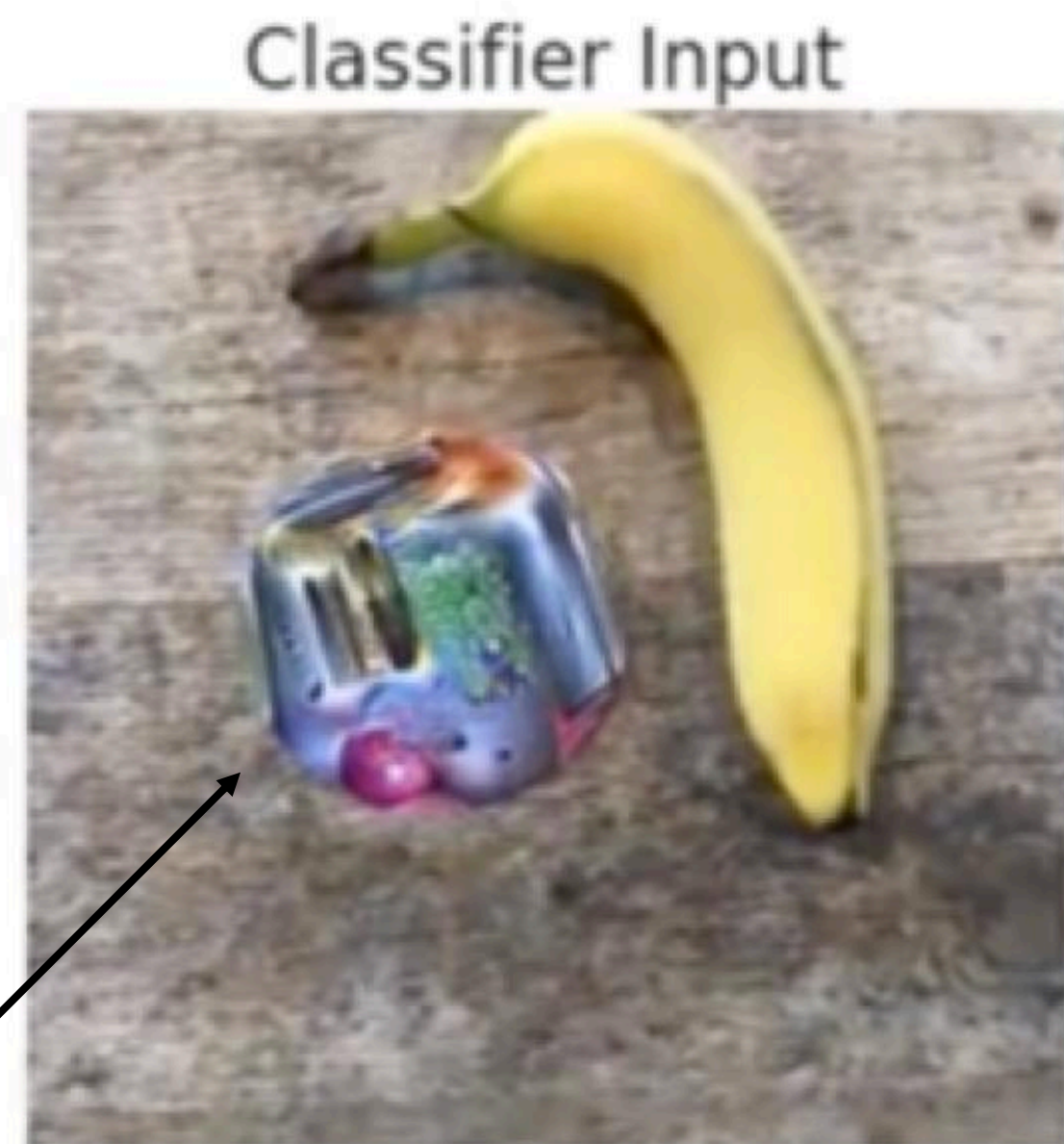
“without the dataset the article is useless”

“okay google, browse to evil.com”

Physical Attacks



Physical Attacks

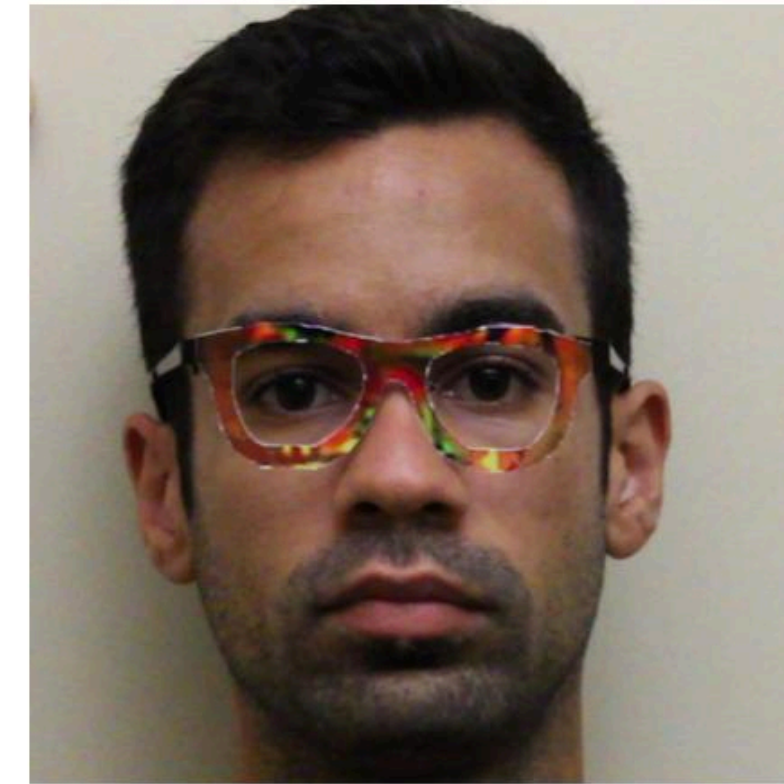


A sticker

Physical Attacks



Physical Attacks



Adversarial Examples in NLP

Article: Super Bowl 50

Paragraph: *“Peyton Manning became the first quarterback ever to lead two different teams to multiple Super Bowls. He is also the oldest quarterback ever to play in a Super Bowl at age 39. The past record was held by John Elway, who led the Broncos to victory in Super Bowl XXXIII at age 38 and is currently Denver’s Executive Vice President of Football Operations and General Manager. Quarterback Jeff Dean had jersey number 37 in Champ Bowl XXXIV.”*

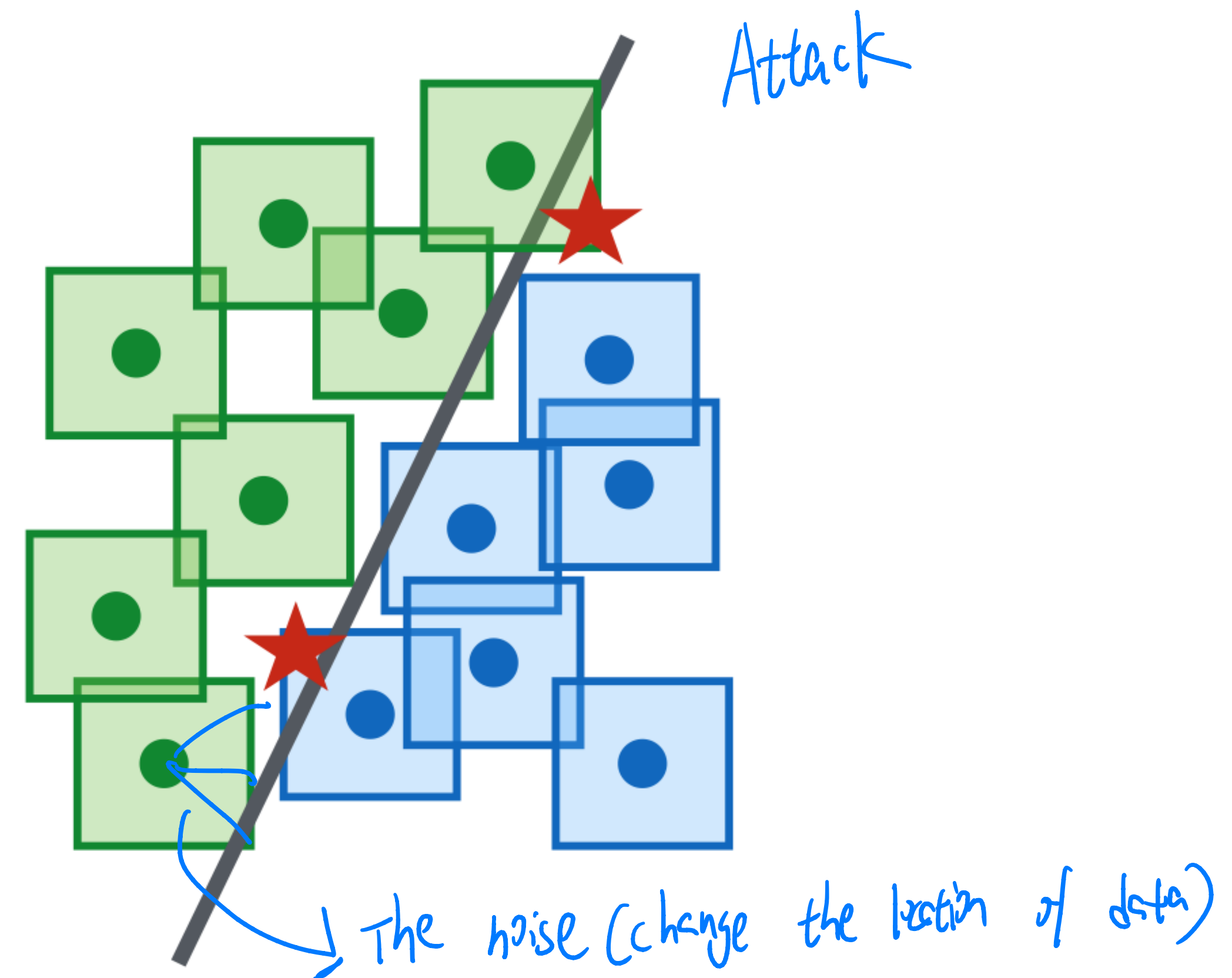
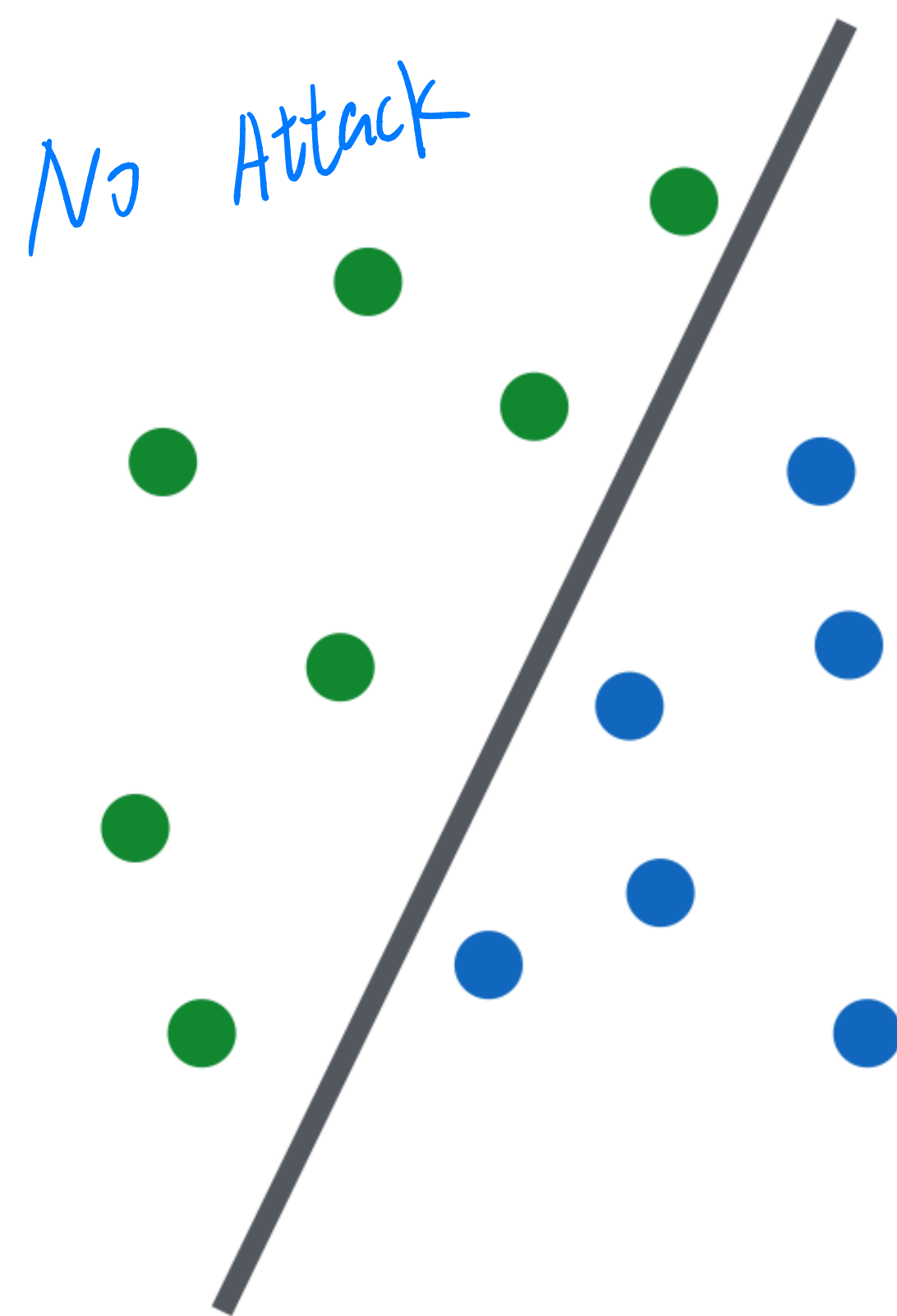
Question: *“What is the name of the quarterback who was 38 in Super Bowl XXXIII?”*

Original Prediction: John Elway

Prediction under adversary: Jeff Dean

Test-time Attack

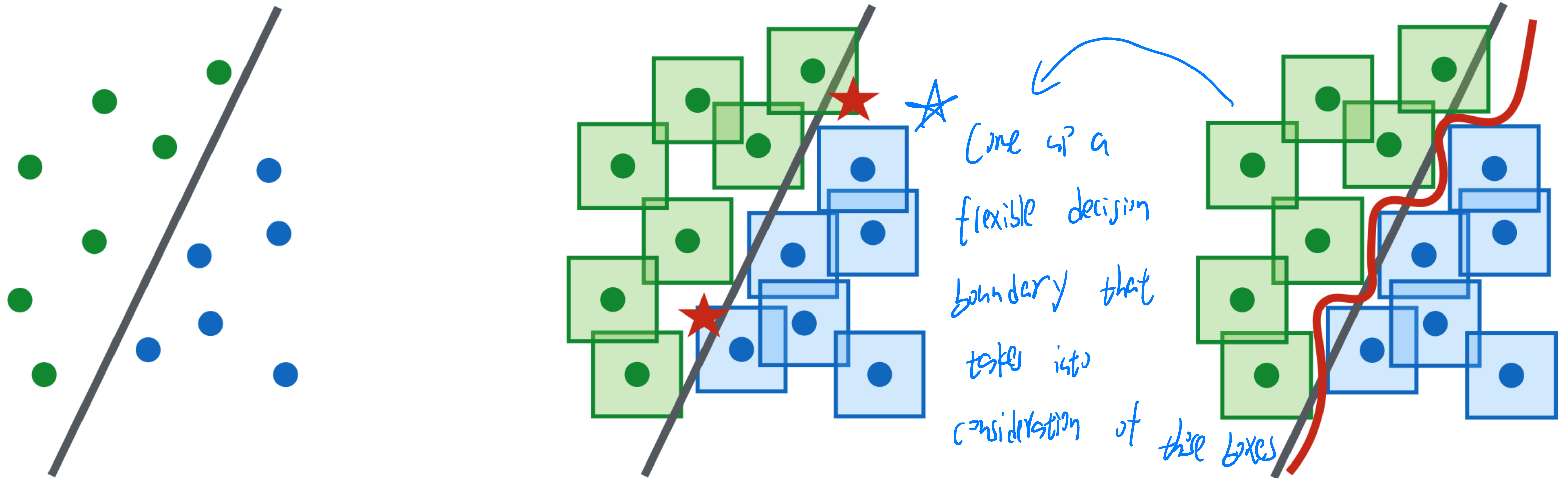
$$\max_{\delta \in \Delta} \ell(x + \delta, y, \theta)$$



(One) Defense against Test-time Attack

Adversarial Training

$$\min_{\theta} \mathbb{E}_D \max_{\delta \in \Delta} \ell(x + \delta, y, \theta)$$



Summary

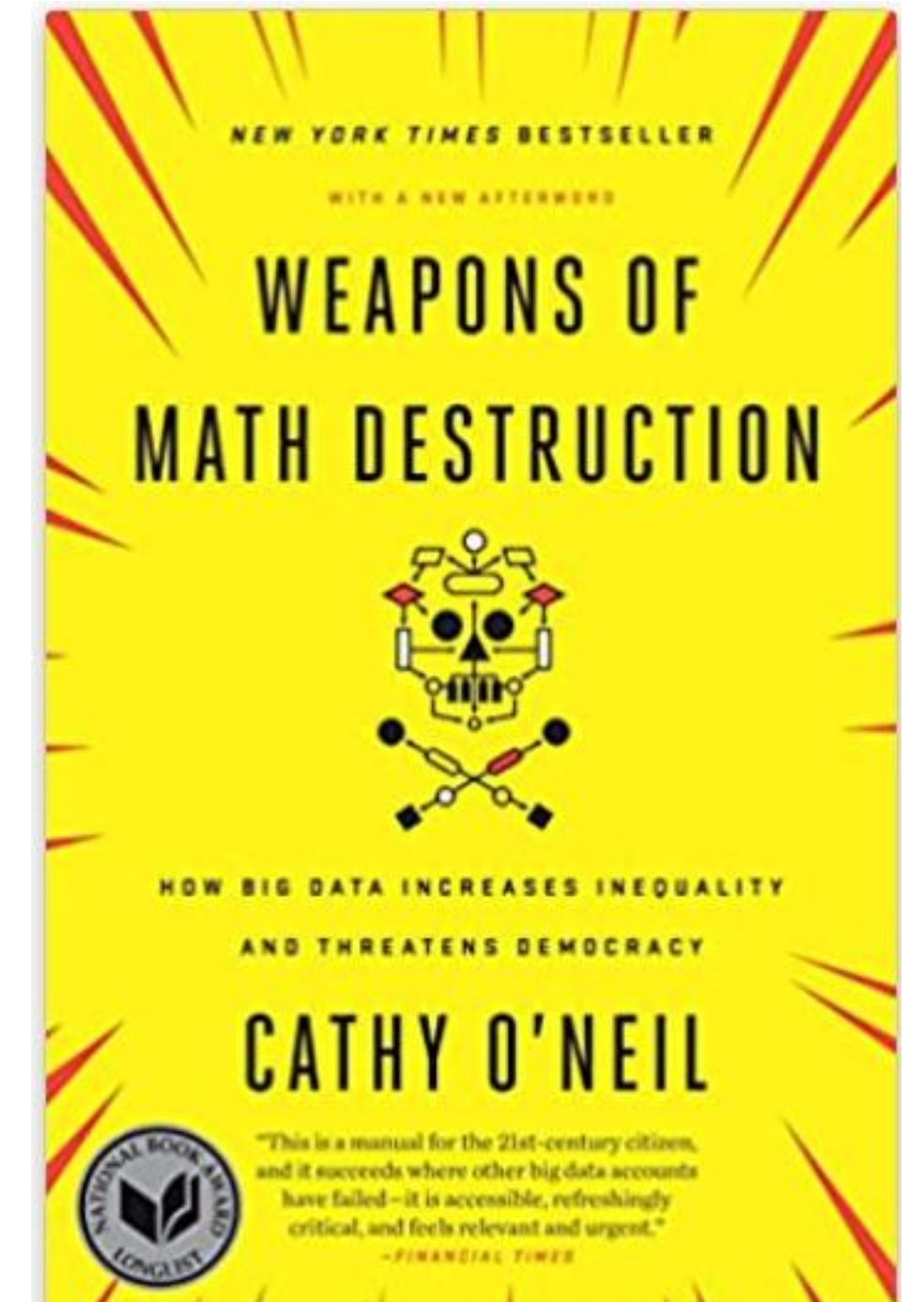
- Bias and Fairness
- Fake content and misinformation
- Privacy
- Adversarial robustness
- **Not covered: value alignment, automation of jobs, equity**
 - **Still important!**

Resources

- **Recommended reading:**

- “Weapons of Math Destruction”
- “Concrete Problems in AI Safety.” Amodei et al.
- “On the Dangers of Stochastic Parrots. Can Language Models be too Big?” Bender et al.

- **Philosophy 244: Introductory Artificial Intelligence (AI) and Data Ethics**



<https://arxiv.org/pdf/1606.06565.pdf>

<https://dl.acm.org/doi/10.1145/3442188.3445922>